	heat capacity calculated **(ii)** would be larger.
9(a)	The root-mean-square (r.m.s.) value of an alternating current is the equivalent value of a steady direct current that will dissipate heat at the same average rate as the alternating current in a given resistor.
9(b)(i)	From $V = 240\sin 377t$, $2\pi f = 377$ $f = \dfrac{377}{2\pi} = 60\text{ Hz}$

9(b)(ii)	$V_{\text{r.m.s.}} = \dfrac{240}{\sqrt{2}} = 169.71\text{ V}$ $P = \dfrac{V_{\text{r.m.s.}}{}^2}{R}$ $= \dfrac{169.71^2}{38}$ $= 757.9$ $\approx 758\text{ W}$
9(c)(i)1.	Faraday's law of electromagnetic induction states that the electromotive force (e.m.f.) induced in a circuit is directly proportional to the rate of change of magnetic flux linkage or rate of cutting of magnetic flux.
9(c)(i)2.	Transformer works by the process of electromagnetic induction. The alternating current in the primary coil produces an alternating magnetic field. This magnetic field is concentrated within the high permeability soft iron core, which causes a large rate of change of magnetic flux linkage in the secondary coil. According to Faraday's law of electromagnetic induction, the changing magnetic flux linkage in the secondary coil induces an e.m.f. in it. Since the rate of change of magnetic flux linkage alternates due to the alternating primary current, the induced e.m.f. in the secondary coil will also alternate with the same frequency as that of the primary current.
9(c)(ii) 1.	An ideal transformer is free from any power loss, such as no core losses, no ohmic resistance in the windings and no leakage of magnetic flux. Hence, the output power $I_S V_S$ is equal to the input power $I_P V_P$.
9(c)(ii) 2.	Using $\dfrac{N_S}{N_P} = \dfrac{V_S}{V_P}$, $N_S = \dfrac{V_S}{V_P} \times N_P$ $= \dfrac{12}{169.71} \times 5000$ $= 353.5$ ≈ 354
9(c)(iii) 1.	Soft iron has a high magnetic permeability. Hence, in order to get the same magnitude of magnetic flux as an air core transformer, soft iron core transformer can be smaller in size. In order to obtain induced e.m.f. at the output of the transformer, there is a need to use high frequency alternating current which results in high frequency reversals in the direction of the magnetic field in the transformer core. Using soft iron would allow the direction of the magnetic flux in the core to be changed easily, lessening the amount of energy lost and increasing the efficiency of the transformer.

9(c)(iii) 2.	The changing magnetic flux linkage within the soft iron core induces eddy currents which cause heat loss. Lamination inserts thin insulating coatings between the soft iron plates to break the flow of eddy current between the plates but do not affect the flow of magnetic flux. This limit the eddy current to flow within each plate, hence reducing power loss during to eddy currents significantly.